

On Optimal Unemployment Compensation*

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Abstract

This paper analyzes the design of an optimal unemployment compensation scheme. The problem is modeled as a dynamic principal-agent relationship between a risk neutral planner (the principal) and risk averse workers (the agents), where the planner's inability to observe workers' job-search efforts creates a moral hazard problem. To design an implementable scheme, we require that each agent is guaranteed a minimum level of expected discounted utility, regardless of his past history. In contrast with previous studies, we find that the *optimal* contract is quite close to *actual* unemployment compensation schemes, both qualitatively and quantitatively.

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1 Introduction

Unemployment compensation programs are an important ingredient of social welfare policies in developed economies. These programs have been widely criticized because of the adverse effects they can have on worker's incentives to search for a new job. This criticism has stimulated extensive research into optimal insurance schemes that take these perverse effects into account.

A series of papers use the dynamic moral hazard model to analyze the trade-off between (unemployment) insurance and (search) incentives. The key feature of the environment is that the probability of finding a new job (or a wage offer above the agent's reservation value) depends on the search effort made by the agent, which is not observed by the planner. In their seminal work on unemployment insurance, Shavell and Weiss (1979) establish that, because of moral hazard, benefits must decrease throughout the unemployment spell, approaching zero in the limit. Usami (1983) considers a dynamic model with moral hazard, where the probability of reemployment conditional on search may depend on the employment history. In an influential paper, Hopenhayn and Nicolini (1997a) (HN hereafter), extend the analysis of Shavell and Weiss, increasing the number of policy instruments available to the government. Together with the sequence of benefits paid to the unemployed workers, they introduce the possibility of contingent wage taxes after reemployment.

All these models confirm the decreasing benefits result of Shavell and Weiss, a characteristic that seems to be shared by most existing schemes. However, the optimal unemployment insurance programs derived from the standard dynamic moral hazard model, such as the one proposed by Hopenhayn and Nicolini, also have certain features that contrast with what we observe in the real world. First, the optimal contract implies that benefit payments decrease slowly with time and never stop decreasing. Second, the optimal program imposes a tax on the wage the worker receives when he finds a job, which is increasing in the length of worker's previous unemployment spell. In contrast, as reported in Figure 1 and Table 1, most OECD countries' unemployment compensation schemes pay a decreasing level of benefits for a fixed period of time and a constant minimum level thereafter.¹ Moreover, none of the observed schemes presents duration-dependent wage taxes. Third (partially given by the combination of the first two points), the difference between the after-tax wage and the unemployment-benefit in HN is rather small compared with existing unemployment compensation schemes.²

[Figure 1 here]

¹Most OECD countries have two types of unemployment benefits. The unemployment insurance system and the assistance system. The unemployment assistance system grants supplementary income to workers who have exhausted the insurance system benefits or who do not qualify for receiving them. Unemployment insurance systems in EFTA countries have shorter duration periods than other countries. For EFTA countries, the average duration is around 0.5/1 year, whereas, in many other OECD countries the transfers may continue for an indefinite number of periods. The behavior of the unemployment benefit payments always presents a downward jump after a short period of time (from 6 months to 3 years) and an almost fixed level of transfers thereafter (see Reissert et al. (1994), Kalish et al. (1998), OECD (1994), Chapter 1 in Layard et al. (1994) and Blöndal et al. (1995)).

²An example of the optimal scheme proposed by Hopenhayn and Nicolini is reproduced in Figure 6 (dotted lines), which will be briefly commented below in this section.

The discrepancy between *optimal* and *actual* programs has obvious implications for government policies. For example, HN argue that by switching from the existing policy to the their optimal transfer scheme, the US government could save approximately between 15 to 30% of the overall spending on unemployment compensations. Hopenhayn and Nicolini (1997b) find costs reduction of the same magnitude for Spain. In this paper we show that most of the discrepancies between the optimal schemes such as the one proposed by Hopenhayn and Nicolini and the ones implemented throughout most developed countries arise from the assumption in the latter that the planner can, and will, inflict infinite punishments on workers. In particular, we first show that the optimal contract derived from the standard dynamic moral hazard model implies a weaker form of what is known as the “immiserization result”:³ if the worker’s utility function is unbounded below then efficiency requires that worker’s expected discounted utility falls, with positive probability, below any arbitrary negative level. We will then argue that relaxing this assumption leads to an optimal contract which is quite close to existing unemployment compensation schemes.

The infinite punishments assumption is questionable in some circumstances for at least two main reasons. First, it may be impossible for the planner to enforce, *ex-post*, such punitive plans, for example because these would imply excessive social conflict costs. Second, workers may have ways of opting out of the insurance scheme. This could be the case if long-term unemployed workers could find a job in the informal economy, if there are migration possibilities, or if the worker can simply leave the program opting for self-sufficiency or relying on family support.⁴

In this paper, we consider the possibility that, when designing the optimal program, the planner must respect a lower bound on the expected discounted utility that the agent can have *ex-post* regardless of the previous history, and we characterize the optimal transfer scheme.

The main *qualitative* features can be summarized as follows. First, at the beginning of the unemployment duration, benefit payments decrease much more steeply with time compared with the benefits implied by the unrestricted model without minimum utility bound, and involve a “last jump” to a minimum. We refer to this minimum benefits level as the “long-term unemployment” or “subsistence” level. When the unemployment benefits reach this level, the transfers stay constant during the remaining unemployment spell and the level of payments depends only on the lower bound on the agent’s expected discounted utility. Second, once a long-term unemployed worker finds a job, he pays a *fixed tax* (which can be equal to zero) which does not depend on the length of the long-term unemployment period.

Looking at Figure 6 it is possible to have an intuition our results. Figure 6 displays a parametrized comparison between the optimal unemployment compensation scheme implied by the unrestricted model of HN (dotted lines), and the restricted one - with utility bounds - that we propose in this paper (in solid lines). In both cases, the upper

³See Thomas and Worral (1990).

⁴An additional reason why the infinite punishment assumption is questionable comes from Atkeson and Lucas (1995). They use a dynastic interpretation of the infinite horizon model. Under this interpretation, an arbitrarily low utility level after some date means that a huge burden is imposed on future generations since ancestors could, and would, sell the consumption of their heirs without limit.

lines represent the net wage w_t - i.e. the after-tax wage a worker should receive once he finds a job - as a function of the unemployment period, in percentage terms of the gross wage. The lower lines represent the replacement ratios for benefits b_t . In the unrestricted case, the two lines are very close to each other, since the use of dynamic incentives allows the planner to reduce the within-period difference between unemployment benefits and after-tax wage, by back-loading punishments. In the restricted case, when the lower bound constraint on worker's expected discounted utility is binding, the scope for dynamic incentives vanishes (since b_t cannot be lowered anymore), and the difference between the unemployment benefits and the net wage has to be big, and equal to the difference arising in the static problem. Thus, on optimal contract implies that the difference between after-tax wage and unemployment benefits increases during unemployment spell, and approaches the "static difference" after finitely many periods. The optimal way to increase the difference between w_t and b_t , is to design a flat net wage schedule (implying a constant wage tax) together with a steeply decreasing unemployment benefits path. Finally, when the unemployment benefits reach their minimum level, the transfers stay constant forever, due to the presence of the minimum utility bound.

A *quantitative* analysis is also performed, using data for the Spanish economy. We find that the actual scheme is not too far from the optimal one. However, according to our model, the Spanish unemployment compensation system is too generous during intermediate periods of unemployment, and it provides benefits payments which are too low for both very-short-term and long-term unemployed workers, compared to the optimal ones.

A minimum bound on expected discounted utility has already been introduced by other authors. Atkeson and Lucas (1995) characterize the optimal contract in a pure adverse-selection setup with temporary (one-period) job offers. They are mainly interested in income distribution, and their approach is closely related to that in Hansen and Imrohoroglu (1992), where the goal is to quantify the welfare effects on unemployment insurance in general equilibrium. Phelan (1995) modifies the repeated insurance problem with adverse-selection of Green (1987) and Thomas and Worrall (1990) and shows the existence of a non degenerate long-run distribution of consumption. Phelan (1993) analyzes a repeated moral hazard problem between firms and workers and shows that (efficiency) wages should increase with tenure. Finally, Wang and Williamson (1996) provide calibrated dynamic OLG models with moral hazard associated with search effort and job retention. Following Phelan (1994), they assume that each new labor-force entrant obtains a prespecified level of ex ante utility.

The literature on optimal unemployment insurance is relatively new, yet quite extended. However, most of the remaining papers address questions and use approaches that cannot be directly related to our own. The interested reader can refer to the recent summary of Karni (1999).

The methodology we use is sometimes called recursive contracts and some references related to our approach are: Abreu, Pearce and Stacchetti (1990), Fudenberg et al. (1990), Spear and Srivastava (1988), Phelan and Townsend (1991).

The paper is organized as follows. In Section 2 we introduce the recursive formulation for the dynamic moral-hazard model. In Section 3 we present the basic unrestricted version of the model and derive the closed form solution. In Section 4 we present the

restricted model with utility bounds, we characterize the optimal contract in detail and we compare it both with the unrestricted model of HN and with the existing programs. Section 5 concludes.

2 Model

We consider a search model with informational problems and our starting point is HN. Suppose that a risk-neutral planner must design an optimal unemployment compensation scheme for a risk-averse worker. In any given period, the worker can be either employed or unemployed. Jobs are permanent.⁵ If he is employed, then he earns a gross wage S forever. If the worker is unemployed, he can engage in costly job-search, and the higher the search-effort a , the higher his probability $P(a)$ of being employed in the next period. The timing of the model, in the unemployment state, is reported in Figure 2.

[Figure 2 here]

The optimal unemployment compensation would be straightforward if the planner could observe the search effort. In this case, the worker would be fully insured against employment risk, would enjoy constant consumption and it would be possible to implement the action that maximizes total welfare, i.e. the *first best* action.

Suppose however that the planner cannot observe a . In this case, unemployment benefits are not paid only as insurance, but must also play the role of giving incentives for search.

The problem can be formalized using the following contractual terminology. At time zero, the principal-planner offers a contract $\mathcal{W}$ to the agent. This contract is optimal in the sense that it guarantees an initial utility level U_0 to the worker, minimizing the expected discounted value of the net transfers to the agent. In particular, for each date t the contract $\mathcal{W}$ specifies net transfers c_t and a recommended action a_t as a function of the realized history $h^t = (h(0), \dots, h(t))$. The history h^t is a vector of zeros and ones, where we denote $h(t) = 1$ if the worker is employed at the *beginning* of period t and $h(t) = 0$ if he is unemployed. Thus, the principal-planner designs the unemployment insurance scheme $\mathcal{W}$ choosing optimally a pair of functions $\{c_t(h^t), a_t(h^t)\}_{t=0}^{\infty}$ taking into account the moral hazard problem in the unemployment state. The planner knows that the worker orders the implied stochastic processes of consumption and search efforts $\{c_t, a_t\}$ according to

$$\mathbf{E} \sum_{t=0}^{\infty} \beta^t [u(c_t) - C(a_t)], \quad (1)$$

where the probability distribution, with respect to which the expectation in (1) is taken, depends on the implied sequences of actions. We assume that the utility function $u(\cdot)$ is strictly increasing and strictly concave while the effort cost function $C(\cdot)$ is assumed to be strictly increasing and convex. We denote by $\beta \in (0, 1)$ the intertemporal discount

⁵In the previous version of the paper we allow for a positive firing probability. The assumption of permanent jobs does not changes any of our qualitative results. Moreover, also HN assume permanent jobs.

factor. We use the symbol b_t to denote the unemployment transfer, i.e. $c_t \equiv b_t$ if $h(t) = 0$ and we use w_t to denote the *net wage* (if the worker is employed), i.e. $c_t \equiv w_t$ if $h(t) = 1$.

Following the recursive contracts literature, the contract can be formulated recursively using current employment situation $h(t)$ and worker's expected discounted utility U_t as state variables.⁶ Consider first the unemployment state case ($h(t) = 0$), and let U be the discounted utility promised to the agent at the beginning of the period. Given a utility level U and present state $h = 0$, recursivity means that the problem can be stated in terms of three functions $[a(U), b(U), U'(0, U, h')]$ determining the current action $a_t = a(U_t)$, the current net transfer $b_t = b(U_t)$, and a promised future utility $U_{t+1} = U'(0, U_t, h(t+1))$, $h(t+1) \in \{0, 1\}$, which is contingent on the search process outcome.

If we define $U^u(0, U) \equiv U'(0, U, 0)$ and $U^e(0, U) \equiv U'(0, U, 1)$, then the choice of the functions $[a(U), b(U), U^e(0, U), U^u(0, U)]$ must satisfy the following two sets of constraints:

$$U \geq u(b(U)) - C(\hat{a}) + \beta[(1 - P(\hat{a}))U^u(0, U) + P(\hat{a})U^e(0, U)] \quad \forall \hat{a}. \quad (\text{IC0})$$

$$U = u(b(U)) - C(a(U)) + \beta[(1 - P(a(U)))U^u(0, U) + P(a(U))U^e(0, U)] \quad (\text{PK0})$$

Constraint (IC0) is the *incentive compatibility constraint* ensuring the agent is willing to deliver the amount of effort called for in the contract. Equation (PK0) requires the contract to deliver the promised level of discounted utility to the worker, and is called the *promise keeping constraint* and plays the role of law of motion for the state variable U . The equivalent promise keeping equation in the employment state is

$$U = u(w(U)) - l + \beta U^e(1, U) \quad (\text{PK1})$$

where $w(U)$ is the net wage the worker receives after tax ($\tau = S - w$) is paid, and we assume a fixed effort cost of working equal to $l \geq 0$. The starting value U_0 will be given by the time-zero participation constraint of the contract $\mathcal{W}$.⁷

Recursivity uses the fact that, under some conditions, second-best contracts have the property that any continuation of the contract is still *on* the constrained utility possibility frontier. Our unemployment compensation problem has two different Pareto frontiers which are precisely the planner's value functions in the two states. The planner's value function in the unemployment state V_u is defined as follows:

$$\begin{aligned} V_u(U) &= \sup_{a, b, U^u, U^e} -b + \beta[P(a)V_e(U^e) + (1 - P(a))V_u(U^u)] \\ &\text{s.t. (IC0) and (PK0).} \end{aligned}$$

⁶Pavoni (1999) contains a formal proof of this recursivity result.

⁷To clarify further the law of motion interpretation of the promise keeping constraints, let us be some-what redundant in notation and rewrite (PK0) and (PK1) as follows

$$\begin{aligned} U_t^u &= u(b_t) - C(a_t) + \beta[(1 - P(a_t))U_{t+1}^u + P(a_t)U_{t+1}^e] \\ U_t^e &= u(w_t) - l + \beta U_{t+1}^e. \end{aligned}$$

It should now be easier to see that each equation implicitly maps values of the state variable in period t into values of the state variable in period $t + 1$.

The problem in the employment state is much simpler since there are no incentive problems. The associated planner's value function V_e is

$$\begin{aligned} V_e(U) &= \sup_{w, U^e} S - w + \beta V_e(U^e) \\ &s.t. \text{ (PK1)} \end{aligned}$$

3 Unrestricted Case: The Immiserization Result

In what follows we will solve the problem when there are only two actions. We assume $a \in \{0, 1\}$, i.e. the worker can either “search” ($a = 1$) or “not search” ($a = 0$). We define $P(1) = p \in (0, 1)$ and $P(0) = 0$; moreover, we normalize the search costs by setting $C(1) = c > 0$ and $C(0) = 0$. Finally, to make the problem interesting, we assume it is always optimal to induce the agent to supply $a = 1$ while unemployed.⁸

Analogously to the general case, the value functions V_u and V_e for a planner facing respectively an unemployed and an employed worker to whom a level U of utility is promised are defined by

$$V_u(U) = \sup_{b, U^u, U^e} -b + \beta [pV_e(U^e) + (1-p)V_u(U^u)] \quad (2)$$

$$s.t. \quad : \quad U^e \geq U^u + \frac{c}{\beta p} \quad (3)$$

$$U = u(b) - c + \beta [(1-p)U^u + pU^e]. \quad (4)$$

and

$$V_e(U) = \sup_{w, U^e} S - w + \beta V_e(U^e) \quad (5)$$

$$s.t. \quad : \quad U = u(w) - l + \beta U^e \quad (6)$$

where (3) and (4) are the incentive compatibility and promise keeping constraints respectively.

It can be verified directly that when the utility of the agent takes the logarithmic form, i.e. $u(x) = \ln(x)$, $l = c$, a solution to the functional equation (2)-(5) is

$$V_u(U) = \frac{SK}{1-\beta} - \frac{B \exp\{(1-\beta)U\}}{1-\beta} \quad (7)$$

$$V_e(U) = \frac{S}{1-\beta} - \frac{\exp\{c\} \exp\{(1-\beta)U\}}{1-\beta} \quad (8)$$

where $K \in (0, 1)$ and $B \in (\exp\{c\}, \exp\{\frac{c}{p}\})$ are constants.⁹ An added bonus is that the implied policy functions are linear in the space of utilities. In particular, for the

⁸Intuitively, this assumption will always be met for a sufficiently high gross wage level S . Equivalently, as we will see below, the exact condition specifies a range of utilities as a function of the whole space of parameters.

⁹In particular $K = \frac{\beta p}{1-(1-p)\beta}$ and B solves

$$\ln \left[B^{\frac{1}{\beta}} - (1-p)B \right] = \ln(p) + \frac{1-\beta}{\beta} \frac{c}{p} + c.$$

unemployment state, the policy functions are

$$u(b(U)) = (1 - \beta)U + \ln B \quad (9)$$

$$U^u(0, U) = U - \frac{\ln B}{\beta} \quad (10)$$

$$U^e(0, U) = U + \frac{\frac{c}{p} - \ln B}{\beta}. \quad (11)$$

The employment state policy is simply a constant utility $U^e(1, U) = U$ for each period, guaranteed by a within-period net wage equal to

$$u(w(U)) = (1 - \beta)U + c. \quad (12)$$

In Figure 3 we have drawn the unemployment state policy functions. They are straight lines in the space of utilities. Figure 3(b) describes the dynamics of the system (equations (10) and (11)). Incentive compatibility implies that the expected discounted utility decreases if the worker remains unemployed while it increases if the worker finds a job.

[Figure 3 here]

The formal counterpart of the previous discussion is the following proposition.

Proposition 1 *Assume $u(x) = \ln(x)$, $l = c$, and let $L = \frac{\ln(\frac{KS}{B-1})}{1-\beta}$, with K and B defined above. Then (i) the value function for the planner's problem in the unemployment state is given by (7) if $U \leq L$, and $V_u(U) = -\frac{\exp\{(1-\beta)U\}}{1-\beta}$ otherwise and (ii) the planner's employment state value function is given by (8) for any range of utilities.*

In Pavoni (2002a) I prove the equivalent of this proposition for a more general problem.

From Figure 3(b) we can see that if the worker stays unemployed for a sufficiently long period, then his utility reaches an arbitrarily low level. This is the graphical intuition for the next proposition.

Proposition 2 *Fix an arbitrary negative level of utility $\underline{U}$. Then given any initial level of utility U_0 there is a positive probability that the worker's expected discounted utility falls below $\underline{U}$.*

Proof. Let $\Delta = U_0 - \underline{U}$, and $\delta = U - U^u(0, U)$. From (10) we have $\delta = \frac{\ln(B)}{\beta}$ which is independent of the level of the state variable U . So, given $\underline{U}$ and U_0 , there will be a natural number N such that $\Delta \leq N\delta$. Then, if the worker cannot find a job for N subsequent periods, the utility will fall below $\underline{U}$, and this happens with probability $(1 - p)^N > 0$. ■

Thomas and Worrall (1990), in a model with adverse selection and i.i.d. shocks, prove a much stronger result. Namely, they show that if the utility is bounded only above and displays non-increasing absolute risk aversion, then any arbitrary low $\underline{U}$ is reached with probability one. This result is known in the literature as the *immiserization* result.

Remark 1 *HN, in their Proposition 1, show that as long as V_u is concave then $U > U^u(0, U)$. From the Theorem of the Maximum we expect the policy $U^u(0, U)$ to be continuous in U . This would imply that the result of Proposition 2 still holds for any unbounded below agent's utility functions, with an arbitrary number of actions. Indeed, for any two fixed levels $U_0 > \underline{U}$, the closed interval $I = [\underline{U}, U_0]$ is a compact set and $f(U) = U - U^u(0, U)$ would be a continuous function of U . Thus, we could use the positive number $\delta = \min_{U \in I} f(U)$ as we did in Proposition 2 to obtain the desired result.*

4 Restricted Case: Imposing Utility Bounds

A key feature of the present paper is to impose a lower bound $U_{\min}$ on the expected discounted utility that can be assigned to the agent from any date onward. From the simplicity of the constraint, it should be easy to see that the problem remains recursive in a very natural way. The Bellman equation for a planner facing an unemployed worker to whom a level U of utility is promised is now given by

$$\hat{V}_u(U) = \sup_{b, U^u, U^e} -b + \beta [p\hat{V}_e(U^e) + (1-p)\hat{V}_u(U^u)] \quad (13)$$

s.t. (3), (4) and

$$U^e, U^u \geq U_{\min}. \quad (14)$$

The problem in the employment state is analogously defined as follows

$$\hat{V}_e(U) = \sup_{w, U^e} S - w + \beta\hat{V}_e(U^e) \quad (15)$$

s.t. (6) and

$$U^e \geq U_{\min}. \quad (16)$$

Notice that we have simply added the minimum bound constraint to problem (2)-(6).

4.1 Characterization of the Optimal Compensation Scheme

In this section, we study the qualitative characteristics of the optimal contract. We will show that the newly-defined value functions are very well shaped: the value functions are strictly decreasing, strictly concave and continuously differentiable. We will then take advantage of these properties to characterize the optimal contract using first order conditions.

Proposition 3 *The value function $\hat{V}_u$ is strictly decreasing, strictly concave and the policy rules are continuous functions in U .*

The proofs of this section can be found in the Appendix.

The whole idea of the proof is to show that the Bellman Equation (13) define a contraction in the space of bounded, continuous and concave functions.

Proposition 4 *Let $\hat{V}_u(U_0) = -b_0 + \beta [p\hat{V}_e(U_0^e) + (1-p)\hat{V}_u(U_0^u)]$ with U_0 in the interior of the effective domain of $\hat{V}_u$ with b_0 in the interior of the domain of the agent's utility function $u(\cdot)$. If $u(\cdot)$ is differentiable then the value function $\hat{V}_u$ is continuously differentiable at any such U_0 and*

$$\hat{V}'_u(U_0) = -\frac{1}{u'(b_0)}.$$

Under the same conditions we also have

$$\hat{V}'_e(U_0) = -\frac{1}{u'(w_0)}.$$

The idea of the proof is quite simple. As in many concave dynamic models, we show differentiability by adapting the Benveniste and Scheinkman (1979) Lemma to our case.¹⁰

The first order conditions in the unemployment state are:

$$\hat{V}'_u(U) = -\frac{1}{u'(b(U))} \quad (17)$$

$$\hat{V}'_e(U^e(0, U)) = -\frac{1}{u'(b(U))} - \mu \quad \mu \geq 0 \quad (18)$$

$$\hat{V}'_u(U^u(0, U)) \leq -\frac{1}{u'(b(U))} + \frac{p}{1-p}\mu \quad (19)$$

$$\hat{V}'_u(U) \geq [p\hat{V}'_e(U^e(0, U)) + (1-p)\hat{V}'_u(U^u(0, U))]; \quad (20)$$

if $U^u(0, U) > U_{\min}$ then both (19) and (20) are satisfied with equality;

together with (3), (4), (14). Condition (18) is an equality because from (3) it is easy to see that the lower bound constraint can be binding only for U^u . In (17) we report the envelope condition. Moreover, note that if (4) is a strict inequality then $\mu = 0$. Finally, note that we derived condition (20) from (18) and (19). The optimality conditions for the employment state are:

$$\hat{V}'_e(U) = -\frac{1}{u'(w(U))} \quad (21)$$

$$\hat{V}'_e(U) \geq \hat{V}'_e(U^e(1, U)); \quad (22)$$

if $U^e(1, U) > U_{\min}$ then (22) is satisfied with equality.

Lemma 1 *The incentive compatibility constraint (3) is always binding at the optimum.*

The idea is the following. As long as $U^e > U^u$ implies a net wage payment higher than the benefits (and this is true as long as $l \geq 0$), then this is costly for the planner, who thus tends to minimize this difference.¹¹

¹⁰For a more general characterization of the dynamic moral hazard problem's value function, see Pavoni (2002b). Another important reference is Phelan and Townsend (1991). They allow for random payments and prove (weak) concavity under quite general conditions.

¹¹In the Appendix, we derived the minimal conditions on l for Lemma 1 to hold. The condition $l \geq 0$ is indeed a sufficient one. A necessary and sufficient condition is $l \geq -\frac{1-\beta}{\beta p}c$. When this condition is not satisfied, the problem becomes less interesting. If l is negative the worker receives a utility "bonus" in case of employment. When this bonus is sufficiently high the search incentives are no longer a problem, and the planner can design an optimal contract which fully insures the agent.

We are now ready to characterize the compensation scheme. In the next proposition we consider first the interior case.

Proposition 5 *For each $U \geq U_{\min}$ such that $U^u(0, U) > U_{\min}$ the policy functions satisfy: (i) for $i = u, e$ and $h \in \{0, 1\}$, $U^i(h, U)$ is increasing in U ; (ii) $b(U)$ and $w(U)$ are strictly increasing; (iii) $U^e(1, U) = U$; (iv) $U > U^u(0, U)$ and $U^e(0, U) > U^u(0, U)$.*

The monotonicity results in (i) and (ii) are intuitive. (iii) implies that the worker is fully insured when employment. This is intuitive since there are no informational restrictions in this state. Finally, for later use note that (ii) together with $U > U^u(0, U)$ in (iv) imply that both b and w are (weakly) decreasing in the unemployment spell. Below we will also describe graphically the result of Proposition 5. However, let us first characterize the optimal transfer scheme when $U^u(0, U) = U_{\min}$.

Proposition 6 (i) *There exists a $U^* > U_{\min}$ such that $U^u(0, U) = U_{\min}$ for every U that belongs to the closed interval $[U_{\min}, U^*]$. (ii) Moreover, if $U \in [U_{\min}, U^*]$ then the within period utility $u(b(U))$ varies one-to-one with U and the promised expected discounted utility $U^e(0, U)$ is constant and equal to $U_{\min} + \frac{c}{\beta p}$ for the whole set.*

Result (i) is important since it implies that $U_{\min}$ will always be reached in finitely many periods of unemployment. The idea of the result (ii) above is that when $U^u(0, U) = U_{\min}$, then $u(b(U))$ is used to keep the promise and $U^e(0, U)$ is used to satisfy the incentive compatibility constraint.

Description of the Optimal Contract We now use graphs to summarize the results obtained in Propositions 5 and 6, and to describe the main features of our optimal contract. To draw Figures 4 and 5 we use the qualitative features of our scheme and we choose thin lines to draw the unrestricted policies (from the closed form solution) and thick lines for the restricted case. As Proposition 5 (i)-(ii)-(iii) shows, policies are monotone in U but of course they do not have to be linear. Moreover, restricted policies would typically not overlap the unrestricted case policies. Finally, it might be the case that $U^e(0, U)$ crosses the 45 degree line. However, from Proposition 6 this can only be for sufficiently large utility values: $U > U^*$. None of these features would affect our discussion below.

Figure 4 reports the results of this section about the unemployment state policy functions. As derived in Proposition 6, Figure 4(a) shows that the policies for future utilities are flat near $U_{\min}$. Figure 4(b) depicts the policy $u(b(U))$, which starts at the point $U = U_{\min}$ where the unemployment benefit is equal to

$$b(U_{\min}) = u^{-1}((1 - \beta)U_{\min});$$

hence its slope must locally be equal to one.

The problem in the employment state is very easy. The net wage starts at a level equal to

$$w(U_{\min}) = u^{-1}((1 - \beta)U_{\min} + l),$$

and $w(U)$ increases with U at a slope equal to $(1 - \beta)$.

[Figure 4 here]

To summarize our results, consider Figure 5. In Figure 5(a) we consider a worker who starts unemployed in A . If he remains unemployed, his utility decreases to B and, eventually, reaches the minimum level $U_{\min}$ in C . When $U_{\min}$ is reached, the utility is constant for the whole remaining unemployment spell. Moreover, regardless of the length of the further unemployment duration, once a job is found, the expected discounted utility jumps to D . Figure 5(b) describes the policy functions for the within period utilities $u(b)$ and $u(w)$. From the direct relationship between utility level U and unemployment benefit payment b , the feature described by the unemployment utility U^u applies also to the benefits payment b . That is, $u(b)$ is decreasing until the minimum level is reached (in C). After that point, the scheme pays a constant transfer and, again, once the worker finds a job there will be a jump to D (on the $u(w)$ line since the state has changed) regardless of the number of periods the worker has remained in C .

[Figure 5 here]

What happens if a worker finds a job soon ? From Figure 5 we can see that actually for very-short-term unemployed workers, the wage tax after reemployment is slightly increasing in the length of the previous unemployment spell. Indeed, suppose a worker finds a job in the first period of his unemployment (while he is in A in Figure 5(a)), then his utility will jump to a point that is above D and, from Figure 5(b) we can see that this implies a higher wage (i.e. a lower wage tax), than the one received by a long term unemployed worker (which, as we have just seen, will jump to D when a job is found). However, now the results have changed both quantitatively and qualitatively with respect to the unrestricted case. The qualitative change is that now the wage tax increases slightly only for a finite number of periods. Moreover, Figure 6 shows that the quantitative aspects are also important.

Figure 6 shows and compares the results of a simulation of both our restricted model and the unrestricted one.¹² The base period is one month and the unrestricted replacement ratios are computed according to our closed form solution of the HN model of Section 3. Dotted lines represent the unrestricted case and solid lines the restricted one. In both cases, the upper lines describe the net wage w_t - i.e. the wage (net of taxes) a worker should receive once he finds a job - as a function of the unemployment period, in percentage terms of the gross wage S ; whereas the lower lines represent the replacement ratios for benefits. It is easy to see how both the net wage and the benefit payment in the unrestricted case vary smoothly during the unemployment spell; after 4 years, the wage tax is around 20 per cent. In contrast, our restricted contract gives a net wage that decreases for a while and remains constant afterwards; the implied wage tax ($S - w_t$) never exceeds 5 per cent.

To understand better why (in Figure 6) the benefit payments behavior for the restricted case (lower solid line) is so different from the unrestricted case (lower dotted line), we now give an intuitive explanation. The use of dynamic incentives allows the planner

¹²For expositional purposes we anticipate here some quantitative results. The calibration exercise is discussed in detail in (the next) Section 4.2.

to reduce the within period difference between unemployment benefit and net wage the worker receives after reemployment. However, when the lower bound constraint is binding the scope for dynamic incentive possibilities vanishes, and the difference between the unemployment benefits and the net wage has to be big, and equal to the difference arising in the static problem. Now, if we start from a level of utility for which the lower bound constraint is not binding, then during the unemployment spell the difference between unemployment benefits and net wage has to increase, approaching the “static difference”. There are only two ways of increasing the difference between the two payments: increasing the net wage and decreasing the benefits. Because of consumption smoothing, the net wage cannot increase during the unemployment spell. Thus, the wage can only decrease more slowly (compared with the unrestricted model) for a while, and remain constant thereafter. On the other hand, the benefits have to decrease much more rapidly: one reason is the usual incentive provision, as in HN, the second reason is the mentioned need of increasing the difference between the unemployment benefits and the net wage during the unemployment period. Thus, benefits decrease steeply for a finite number of periods and then become completely flat. In contrast, in the unrestricted case, dynamic incentives are always feasible. As a consequence, both the net wage and the benefit transfer decrease regularly with a much lower slope. During unemployment, benefit payments never stop decreasing, implying, eventually, arbitrarily low levels of utility, as we have shown in Proposition 2.

4.2 Quantitative Analysis

We calibrate our model using data on the Spanish economy. I normalize the gross wage $S = 100$ and the effort costs $c = l = 1$.

We interpret each period as a month so we set the discount factor $\beta = 0.996$ which implies an annual discount rate of 5%. The main departures from HN are the choice of the risk aversion parameter and the estimation of the minimum level of promised utility $U_{\min}$. HN use CRRA-preferences of the form

$$u(x) = \frac{x^{1-\sigma}}{1-\sigma}$$

and choose $\sigma = 0.5$. In our quantitative exercise, we set $\sigma = 1$, i.e. we use the logarithmic utility function. This is mainly done in order to use our closed form solution.¹³ Finally, to estimate the parameter $U_{\min}$ we first set

$$U_{\min} = \frac{u(qS) - c + \beta p U_{work}}{1 - \beta(1 - p)} \quad (23)$$

where q expresses the “social salary” as a function of gross wage S . Thus, $U_{\min}$ corresponds to the expected discounted utility of receiving the social salary forever and always having

¹³However, Attanasio and Weber (1993) use UK cohort data to estimate the intertemporal elasticity of substitution. In our specification - given we assume CRRA preferences - the results of Attanasio and Weber imply a σ between 1.3 and 1.5. Moreover, Baily (1977), who studies issues similar to those in this paper, argues for setting $\sigma = 1$, as we do.

the chance of getting a job, and paying no taxes. The parameter U_{work} is defined

$$U_{work} = \frac{u(S) - l}{(1 - \beta)},$$

and represents the utility of working forever receiving the gross wage S . Thus, U_{min} is calibrated by setting $q = 0.20$.¹⁴ Moreover, note that $u(S) = \ln(100) = 4.6$ so the cost $c = l = 1$ is between one fourth and one fifth of the wage utility, and $U_{work} = 900$.

In Figure 6, we report the simulated replacement ratios for the our restricted model, and we compare them with the ones obtained using the closed form solution of the unrestricted model of HN. We use $p = 0.15$ and the starting value $U_0 \in (U_{min}, U_{work})$ is computed - partially following HN - according to the existing Spanish scheme.¹⁵ We have just commented on Figure 6 in Section 4.1.

[Figure 6 here]

The benchmark level used for p is an intermediate one. Bover et al. (1997) emphasize the high degree of (both observable and unobservable) heterogeneity between workers' hazard rates in the Spanish labor market.¹⁶ Thus, in Figure 7, we investigate how the optimal transfer scheme changes with p . We consider four possible values for $p - 0.08$,

¹⁴The amount of the non-contributive assistance level of transfers, is means-tested, varies across different Autonomous Communities between 30,000 to 45,000 pesetas (\$150/180), and there is no fixed duration (See Lopez (1996)). For example, in Catalonia, the transfer is called PIRMI (Programa interdepartamental de la renta mínima d'inserció) and, in 1998, the monthly payment was approximately 40,000 pesetas (\$160). The Bulletin of Labor Statistics (1999) reports as 300,000 pesetas (\$1,200) per month the 1998 average wage in non-agricultural activities. We estimated the parameter q following the common assumption that workers subject to severe unemployment risk face a wage that is two thirds of the average national wage.

¹⁵Given we normalized the gross wage S to 100, the current insurance system can be represented by a contract that has not taxes or transfers when unemployed ($w = S$), and pays a first benefit level b^1 of 70 for the first six months of unemployment, from the 7th to the 24th month the benefit level b^2 is set equal to 60, and from the 25th onward we assume the worker receives an assistance level of benefits $b^3 = 20$. The corresponding expected discounted utility value U_0 for an unemployed worker, can be calculated backward as follows. From period $T = 25$ onward the worker's problem is stationary. The unemployment benefits are at their minimum level, the worker will be searching for a job, which will be found with probability p . Given that jobs are permanent, when a worker finds a job his lifetime utility is U_{work} . Thus, the value of his expected discounted utility U_T can be computed as follows

$$U_T = \frac{u(b^3) - c + \beta p U_{work}}{1 - \beta(1 - p)} = U_{min},$$

where $b^3 = qS = 20$ is the non-contributive assistance level of unemployment benefits. For any $0 \leq t \leq T = 25$ we can now define the value U_{T-t} recursively by

$$U_{T-t} = u(b_t) - c + \beta [p U_{work} + (1 - p) U_{T-(t-1)}],$$

where the period t benefit level $b_t \in \{b^1, b^2, b^3\}$ is computed according to the three steps scheme described above.

¹⁶Bover et al. find that the average monthly probabilities of exit from unemployment vary across Sector (between 0.11 and 0.17), Age (between 0.08 and 0.15) and the Business Cycle (between 0.12 and 0.18). They also estimate the hazard rate function controlling for unobservable heterogeneity.

0.10, 0.15 and 0.20 —. Figure 7 shows that the first order effect of a change in p is mainly on the steepness of the payments across time. For higher levels of p , the benefits payments decrease more rapidly toward a common minimum replacement ratio level of approximately 35%. This simulation exercise has an immediate policy implication. The optimal unemployment compensation scheme should be a function of each *observable* variable (such as age, labor market tightness, geographical region or industrial sector) that implies a different probability of re-employment. In contrast, the net wage payments are not very sensitive with respect to changes in this parameter.

In Figures 8 and 9 we summarize the results of our sensitivity analysis for different minimum bounds $U_{\min}$ and different starting utilities U_0 , respectively. The results present the same degree of robustness of the results in Figure 7. It may seem surprising that changes in $U_{\min}$ do not too much affect the long-term replacement ratio (which remains approximately 35%). However, if we look carefully at Figure 8, we can see that the optimal contract guarantees higher minimum utility levels, mainly by reducing the wage tax imposed on long-term unemployed workers. Since jobs are permanent this effect is quantitatively important. Moreover, the wealth effect caused by a higher $U_{\min}$ tightens the incentive constraint and induces lower benefit payments, partially offsetting the initial (increasing) one.

[Figures 8 and 9 here]

In Figures 10 and 11, we compare our simulations with the Spanish compensation scheme (the solid step-shaped line).¹⁷ The model seems to reproduce most of the qualitative features of the Spanish compensation system, and this confirms the analysis of Section 4.1. From a *quantitative* point of view the Spanish unemployment compensation scheme seems to be too generous during intermediate periods of unemployment, whereas it provides payments which are too low for both the very short-term and the long-term unemployed people with respect to the optimal transfer scheme. It seems clear that part of this discrepancy can be accounted for by introducing positive administrative costs in our model.¹⁸ However, we think that a deeper analysis of these quantitative aspects not only requires more accurate measurements but, more importantly, probably also requires a bigger set of parameters.¹⁹ This is beyond the scope of the present paper and remains for future research.

[Figures 10 and 11 here]

¹⁷In Spain, the replacement ratio is equal to 70% during the first six months of unemployment and 60% thereafter, subject to a floor of 75% of the minimum wage. Benefit duration is one-third of the last job's tenure, with a maximum of two years. The assistance system pays, for up to two years, 75% of the minimum wage to (unemployed) workers with dependants, whose average family income is precisely below that amount. In 1998, the minimum wage was around 70,000 pesetas (\$280) (Guia Laboral 1998 y de Asuntos Sociales (1998)).

¹⁸I am grateful to Gianluca Violante for drawing our attention to this point.

¹⁹In particular, note that we assume permanent jobs and we use a *naive* hazard function. We always allow only for two actions assuming that $P(0) = 0$, and we require time invariance of $P(a)$. See Pavoni (2002b) for a more general case.

5 Conclusions

In this paper, we studied the design of an implementable optimal unemployment compensation scheme. To analyze the trade-off between unemployment insurance and search incentives, we used a dynamic principal-agent relationship between a risk neutral planner and risk averse workers, where the planner's inability to observe workers' job-search efforts induces moral hazard.

Our analysis can be summarized as follows. In the first part of the paper, we compute a closed form solution for the unrestricted dynamic moral hazard model with two actions and logarithmic utility agent. We use this specification to show that if no bound is imposed on the expected discounted utility promised to the agent, the optimal unemployment compensation schemes such as the one proposed by Hopenhayn and Nicolini imply that the worker's utility falls with positive probability below any arbitrarily low level. This is a weaker form of what is known in the recursive contracts literature as the immiserization result. We thus argue for the importance of restricting the planner's contract space. In particular, we require that, when designing the optimal program, the planner must respect a lower bound on the expected discounted utility that the agent can have *ex-post* regardless of the previous history. We found that the introduction of utility bounds in the optimal unemployment compensation designing problem has important normative and policy implications.

First, the optimal restricted compensation scheme is *qualitatively* consistent with the existing ones, meaning that governments are not making such big mistakes as the one highlighted, for example, by Hopenhayn and Nicolini (1997a,b). We also calibrate and simulate our model for the Spanish economy, and we find only minor *quantitative* differences between optimal and existing ones. We think, however, that, in order to carefully evaluate the (sub)optimality of government policies (and to give meaningful policy suggestions), a larger set of parameters is probably required. Pavoni (2002) focuses, for example, on the possibility that the probability of reemployment depends on the duration of previous period of unemployment, and shows that lower bounds on an agent's utility can arise endogenously in the optimal scheme.

Second, we recall that one of the main policy implications of the optimal program proposed by HN is that an optimizing government should impose a tax on the wage the worker receives when he finds a job, which is increasing in the length of the worker's previous unemployment spell. We show that the introduction of utility bounds eliminates this counterfactual characteristic of the optimal unemployment compensation scheme. In the stationary model studied in this paper, a simple consumption smoothing argument implies that wage tax cannot (strictly) decrease during unemployment. However, in Pavoni (2002b) I show that the policy implication for an extended model which accounts for human capital depreciation during the unemployment spell is a wage subsidy, even in an unrestricted model without utility bounds.

An important extension of the model would be to allow the agent to secretly save (and borrow). In Abraham and Pavoni (2002) we study the welfare implications of such possibility, and show how the problem can again be written in recursive form. Unfortunately, we also find that when the agent is allowed to save, the optimal contract does not

uniquely pin down the government transfer scheme.²⁰ In fact, associated to each second-best allocation for consumption (and search effort) there is a continuum of unemployment compensations schemes implementing such allocation. For example, Werning (2002) obtain different UI transfer schemes depending on the “normalizations” he considers to get rid of the mentioned indeterminacy.

Finally, in the extended version of this paper I also consider self-financing schemes with a continuum of agents and steady state equilibria.

6 Appendix

Proof of Proposition 3 The first technical difficulty is to show that the value functions $\hat{V}_i$ are bounded. To this extent we assume there is also an upper bound $U_{\max}$ on the expected discounted utility promised to the agent. We will then show that $U_{\max}$ can always be chosen sufficiently large so that it will never be binding. As a consequence, the contract we derive is actually the solution to the (relaxed) problem where only the lower bound $U_{\min}$ is imposed. It is easy to show that $\hat{V}_e$ is concave. We now prove that also $\hat{V}_u$ is concave. We follow Grossman and Hart (1983) and change variable defining $z \equiv u(b)$. The problem then becomes

$$\begin{aligned} \hat{V}_u(U) &= \sup_{z, U^u, U^e} -u^{-1}(z) + \beta [p\hat{V}_e(U^e) + (1-p)\hat{V}_u(U^u)] \\ \text{s.t.} \quad & z - c + \beta [(1-p)U^u + pU^e] \geq z + \beta U^u \\ U &= z - c + \beta [(1-p)U^u + pU^e] \\ U_{\max} &\geq U^u, U^e \geq U_{\min}. \end{aligned} \tag{24}$$

Notice that (since $u^{-1}(\cdot)$ is convex) the objective function is concave and that the constraints set is convex (linear). Moreover, the problem satisfies all the conditions of Theorems 4.6-4.8 of SLP. In particular, the functional equation (24) is a contraction since it satisfies the Blackwell’s sufficient conditions, and, using the promise keeping constraint, we have $z = U - \beta [(1-p)U^u + pU^e] - c$ so that $-u^{-1}(U - \beta [(1-p)U^u + pU^e] - c)$ is strictly decreasing and strictly concave in U (since u is strictly increasing and strictly concave). We need now to show that $U_{\max}$ is never binding. However, since this last part of the proof is derived from the characteristics of the optimal contract we postpone it after Proposition 5. ■

Proof of Proposition 4 Again, we should concentrate on the differentiability of $\hat{V}_u$. Given the value function is concave we can use Lemma 2 in Benveniste and Scheinkman (1979). For a fixed level of promised utility U_0 we are looking for a differentiable concave function $W(U)$ such that it is well defined in an interval I around U_0 and such that for any $U \in I$ we have $W(U) \leq \hat{V}_u(U)$ and $W(U_0) = \hat{V}_u(U_0)$. We claim that $W(U) = -u^{-1}(U - \beta U_0^u) + \beta [p\hat{V}_e(U_0^e) + (1-p)\hat{V}_u(U_0^u)]$ is the function we are looking for. Indeed, the optimal values U_0^e and U_0^u satisfy both the incentive compatibility and (14). The constraint (4) can always be satisfied by varying the benefit transfer b , so the function W is well defined and we have $W(U) \leq \hat{V}_u(U) \forall U \in I$ as required. From the concavity and differentiability of u , W is

²⁰This unpleasant result is unfortunately true also when savings are observable. See Abraham and Pavoni (2002) Proposition 4.

also concave and differentiable, so $\hat{V}_u$ is differentiable in U_0 and $\hat{V}'_u(U_0) = -\frac{1}{u'(b_0)}$. Since $\hat{V}_u$ is concave then it is continuously differentiable. To show the continuous differentiability of $\hat{V}_e$ is immediate. ■

Proof of Lemma 1 The proof is given by contradiction. First, consider the case $U^u(0, U) > U_{\min}$. Assume (3) is not binding at the optimum for some level of utility U . Then, $\mu = 0$ and from (17), (18) and (19) we have

$$\hat{V}'_e(U^e(0, U)) = \hat{V}'_u(U^u(0, U)) = \hat{V}'_u(U). \quad (25)$$

From the strict concavity of $\hat{V}_u$ we have $U^u(0, U) = U$. Now consider the employment state problem for the utility level $U^e(0, U)$. The envelope conditions (17), (21) we have $b(U) = w(U^e(0, U))$. Use this facts to rewrite the promise keeping constraints in the two states, take the difference, and rearranging terms we obtain

$$1 - \beta(1 - p) [U^e(0, U) - U^u(0, U)] = c - l \geq \frac{1 - \beta(1 - p)}{\beta p} c \quad (26)$$

where the last inequality is obtained using the fact that from the incentive compatibility constraint we have that $U^e(0, U) - U^u(0, U) \geq \frac{c}{\beta p}$. So using the inequality in (26) and rearranging terms we get

$$l \leq -\frac{1 - \beta}{\beta p} c,$$

which is a contradiction since $l \geq 0$ and the right hand side is strictly negative.

When $U^u = U_{\min}$ we can use the same argument as before, if we show that cannot be that both the incentive compatibility constraint is not binding and that (19) is a strict inequality. But if it were true then $\hat{V}'_u(U_{\min}) < \hat{V}'_u(U)$ which from strict concavity implies $U_{\min} > U$ and it is not possible. ■

Proof of Proposition 5 Since $U^u(h, U) > U_{\min}$ we have that (20) and (22) are satisfied with equality. From strict concavity of the value function and Lemma 1 we have (i). The strict concavity of $\hat{V}_i$ and $u(\cdot)$ - using envelope conditions (17) and (21) - implies (ii).

To prove (iii) recall the first order condition in the employment state, allowing for $U^e(1, U) = U_{\min}$ is (22), thus $U^e(1, U) \geq U$ for any U . Moreover assume that $U^e(1, U) > U$, then - since $U \geq U_{\min}$ - the condition (22) has to be satisfied with equality. Strict concavity of $\hat{V}_e$ implies $U^e(1, U) = U$ and we are done.

Result (iv). Since $\mu \geq 0$ (19) implies that $U \geq U^u(0, U)$ moreover $U = U^u(0, U)$ can only be if $\mu = 0$ and we are in the case of equation (25) in the proof of Lemma 1, and this contradict incentive compatibility. That $U^e(0, U) > U^u(0, U)$ follows trivially from incentive compatibility. ■

Final part of the proof of Proposition 3 Consider a starting value $U_0 < \infty$, $U_0 \geq U_{\min}$ for the unemployed agent's lifetime utility. From Proposition 5 we have monotonicity of the policy functions (result (i),(ii) and (iii) above). Moreover from result (iv) we have that $U^u \leq U$. As a consequence, setting $U_{\max} \geq U_0 + \frac{c}{\beta p} > U_{\min}$ would give a never binding constraint. ■

Proof of Proposition 6 (i) First, we argue that $U^u(0, U_{\min}) = U_{\min}$. From (19) together with the envelope condition (17) we obtain

$$\hat{V}'_u(U^u(0, U_{\min})) \leq \hat{V}'_u(U_{\min}) + \frac{p}{1-p}\mu. \quad (27)$$

Then either the inequality is strict (and this implies $U^u(0, U_{\min}) = U_{\min}$ so we are done) or it is an equality. In the latter case, from $\mu \geq 0$ we have $\hat{V}'_u(U^u(0, U_{\min})) \geq \hat{V}'_u(U_{\min})$, which from the strict concavity of $\hat{V}_u$ in turn implies $U^u(0, U_{\min}) \leq U_{\min}$ and again we are done.

Second, we show by contradiction that, when $U = U_{\min}$ condition (27) is a strict inequality. So assume (27) is satisfied with equality, then since $U^u(0, U_{\min}) = U_{\min}$ from strict concavity of $\hat{V}_u$ must be that $\mu = 0$ and this, together with (17) implies

$$-\frac{1}{u'(b)} = \hat{V}'_u(U_{\min}) = \hat{V}'_e\left(U_{\min} + \frac{c}{\beta p}\right). \quad (28)$$

From Lemma 1, using the promise keeping constraint (4) we have

$$U_{\min} = u(b) + \beta \left[(1-p)U_{\min} + p \left(U_{\min} + \frac{c}{\beta p} \right) \right] - c = u(b) + \beta U_{\min},$$

so

$$u(b) = (1-\beta)U_{\min}. \quad (29)$$

Now consider the employment state when $U = U_{\min} + \frac{c}{\beta p}$.

The envelope conditions is $\hat{V}'_e\left(U_{\min} + \frac{c}{\beta p}\right) = -\frac{1}{u'(w)}$, which, from strict concavity of u and (28) gives $w = b$. Moreover, from the first order condition (22) we have $\hat{V}'_e\left(U_{\min} + \frac{c}{\beta p}\right) = \hat{V}'_e(U^e)$ so the policy should be $U^e\left(1, U_{\min} + \frac{c}{\beta p}\right) = U_{\min} + \frac{c}{\beta p}$. The promise keeping constraint implies $U_{\min} + \frac{c}{\beta p} = u(w) + \beta \left(U_{\min} + \frac{c}{\beta p} \right) - c$. From $w = b$ and (29) this we then have $U_{\min} + \frac{c}{\beta p} = (1-\beta)U_{\min} + \beta \left(U_{\min} + \frac{c}{\beta p} \right) - c = U_{\min} + \frac{\beta(1-p)c}{\beta p}$ which gives a contradiction since $\beta < 1$ and $p \geq 0$.

By the monotonicity shown in Proposition 5 the set at which $U^u(0, U) = U_{\min}$ is an interval, by the maximum theorem the point U^* is given by

$$\hat{V}'_u(U^*) = \left[p\hat{V}'_e(U^e(0, U^*)) + (1-p)\hat{V}'_u(U_{\min}) \right],$$

where (20) is satisfied with equality.

(ii) If we use Lemma 1 and rearrange constraints (3) and (4) then we find that the policy functions must satisfy the following two equations:

$$\begin{aligned} u(b(U)) &= U - \beta U^u(0, U) \\ U^e(0, U) &= U^u(0, U) + \frac{c}{\beta p}. \end{aligned}$$

Moreover, for sufficiently low levels of U , the lower bound constraint for U^u is binding and the policy functions satisfy

$$u(b(U)) = U - \beta U_{\min} \quad (30)$$

$$U^e(0, U) = U_{\min} + \frac{c}{\beta p} \quad (31)$$

and we are done. ■

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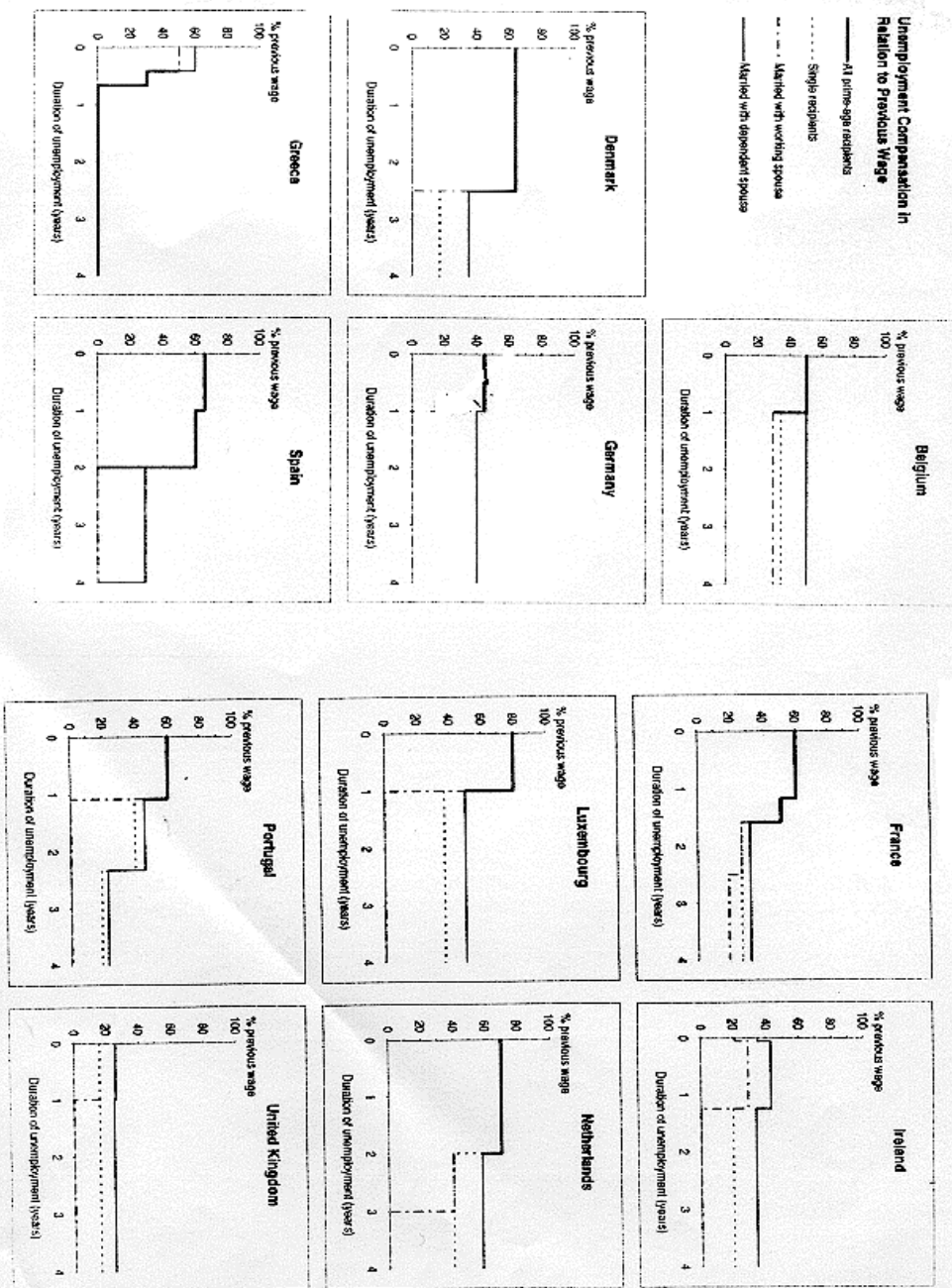


Figure 1: **Some Existing European Unemployment Compensation Schemes.**
Source Reissert, B. and G. Schmid (1994), "Unemployment Compensations and Active Labor Market Policy: The Impact of Unemployment Benefits on Income Security, Work Incentives and Public Policy," in *Labor Market Institutions in Europe*, G. Schmid Ed., Sharpe, US.

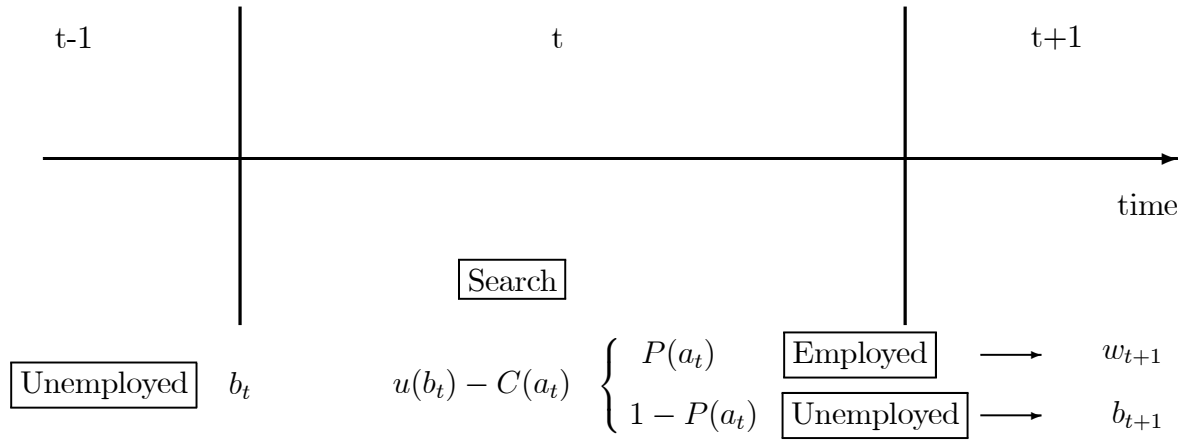


Figure 2: **The timing of the model in the unemployment state.** At the beginning of period t the unemployed worker receives the unemployment benefit b_t and is required to supply the costly job-search effort a_t . This effort affects the probability of being employed at the beginning of the next period $P(a_t)$. If at the beginning of next period the worker is employed he receives a net wage $w_{t+1} = S - \tau_{t+1}$ otherwise, if he still unemployed, the worker receives the unemployment benefit b_{t+1} , and so on.

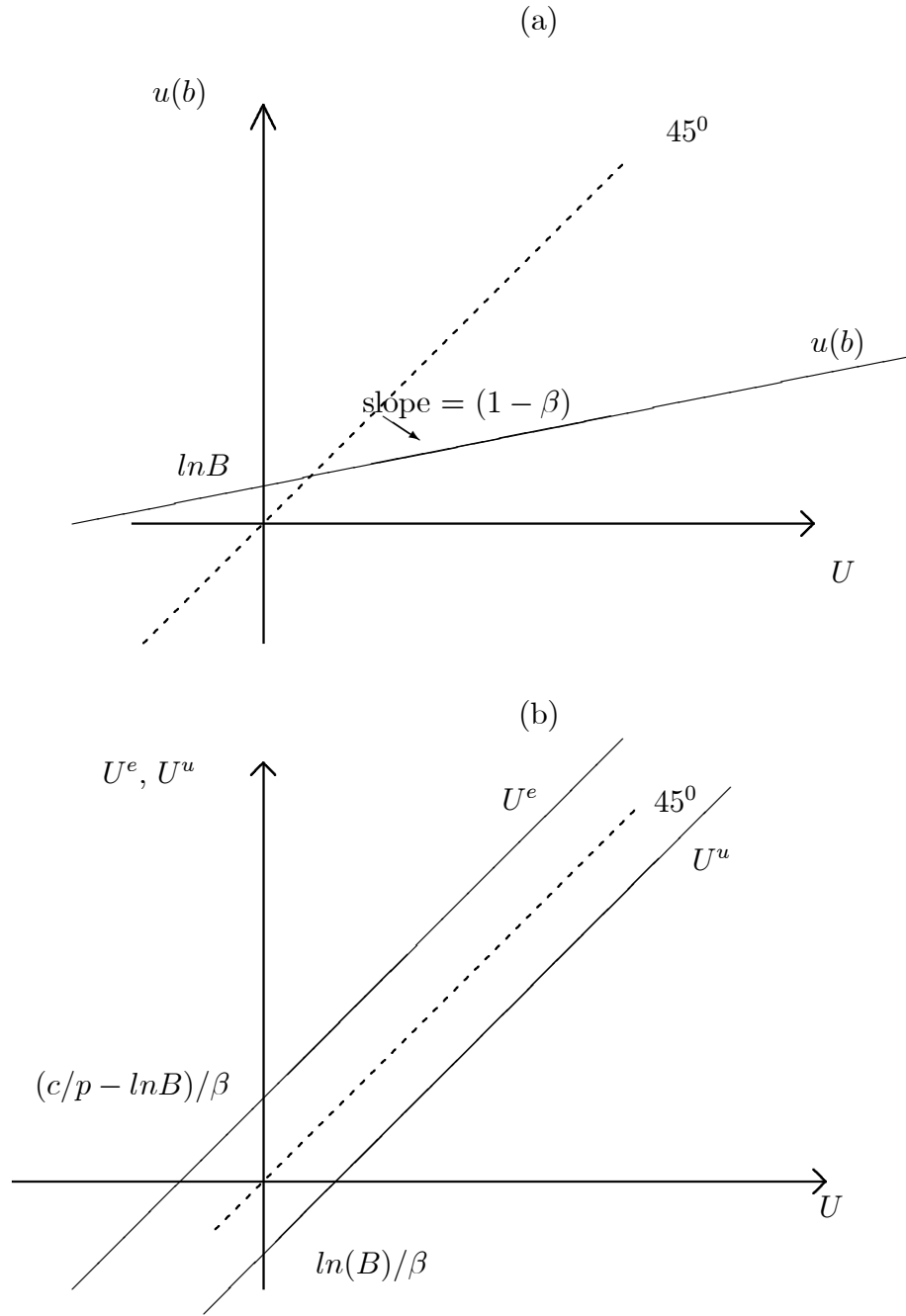


Figure 3: **The policies for the unrestricted model with logarithmic agent's utility function: unemployment state.** Panel(a) reports the policy implied by equation (9). Panel(b) the policies implied by equations (10) and (11).

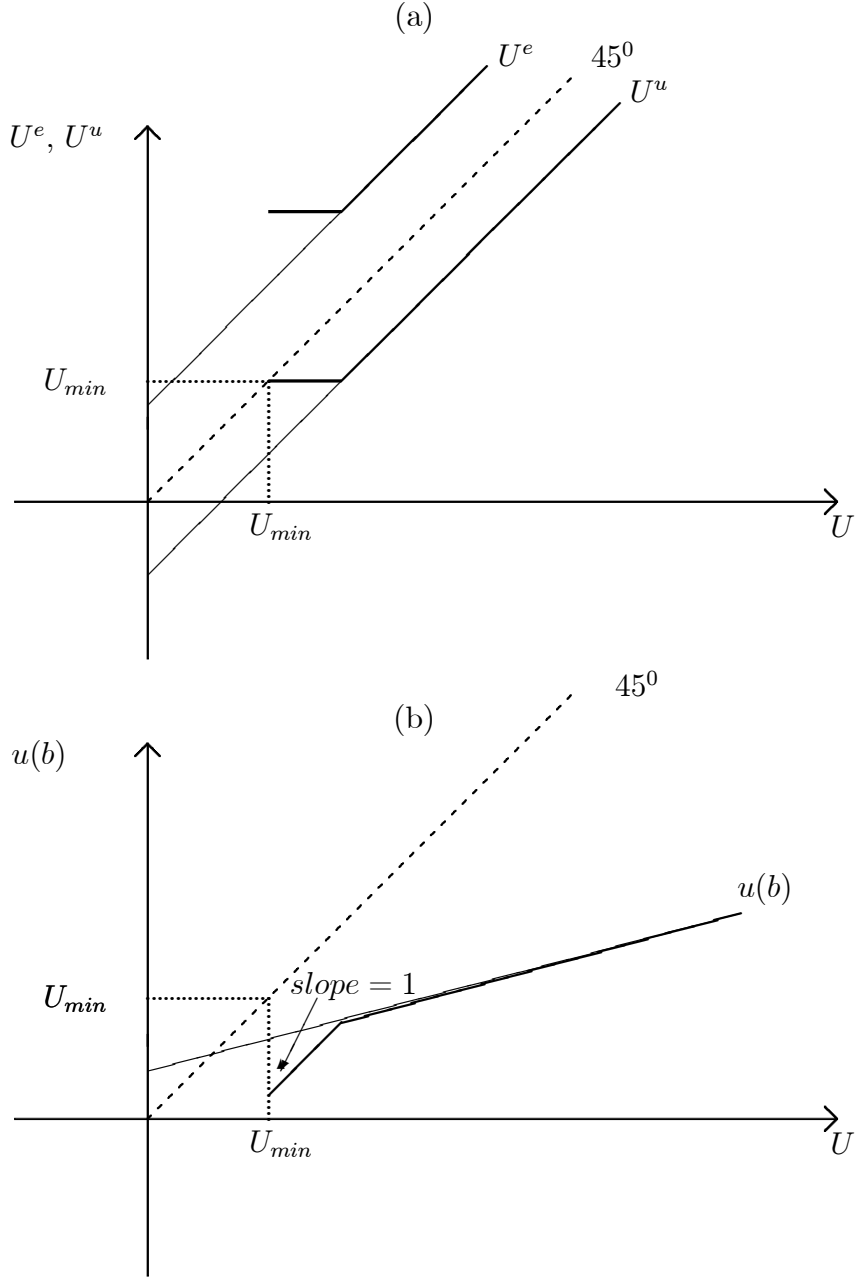


Figure 4: **The policy functions: unemployment state.** In both panels (a) and (b) the thick line reproduce the policies for the restricted scheme, while the thin lines refer to the unrestricted case. Panel(a) shows that the policies for future utilities are flat near U_{min} and tend to the (unrestricted) unitary slope as U increases. Panel(b) depicts the policy $u(b(U))$, which starts at the point $U = U_{min}$ where the unemployment benefit is equal to $b(U_{min}) = u^{-1}((1 - \beta)U_{min})$ and its slope is (locally) equal to one.

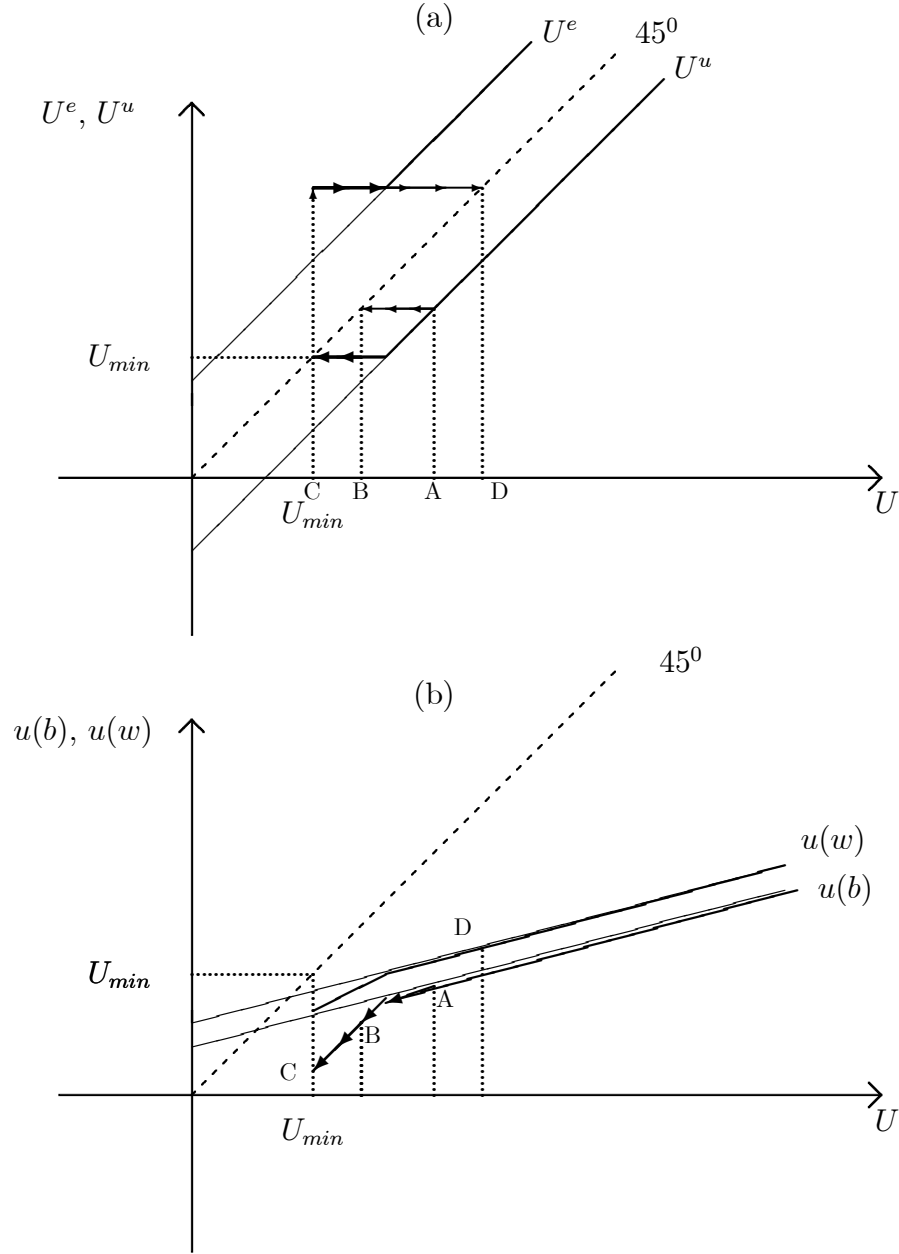


Figure 5: **The long term unemployment behavior.** In panel (a) we consider a worker who starts unemployed in A . If he remains unemployed his utility decreases to B and, eventually, reaches the minimum level U_{min} in C . Once a long-term unemployed finds a job the expected discounted utility jumps to D . Panel (b) reports the within period utilities $u(b)$ and $u(w)$ implied by this utility path.

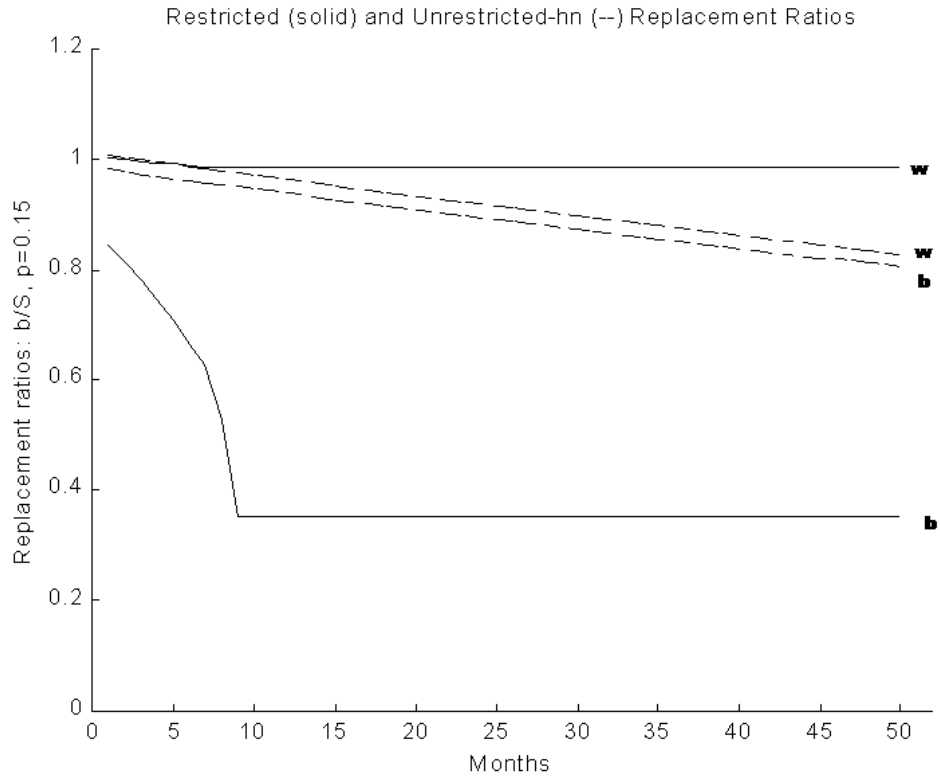


Figure 6: **Comparison between the Restricted and the Unrestricted model.** The dotted lines reproduce the simulation results for the benchmark unrestricted model of HN. The solid lines reproduce our simulation results for the restricted model with utility bound. Lower lines represent benefit payments while upper lines represent after-tax wage payments, as a function of the unemployment duration. The horizontal axis represents unemployment duration (in months). The vertical axis represents replacement ratios.

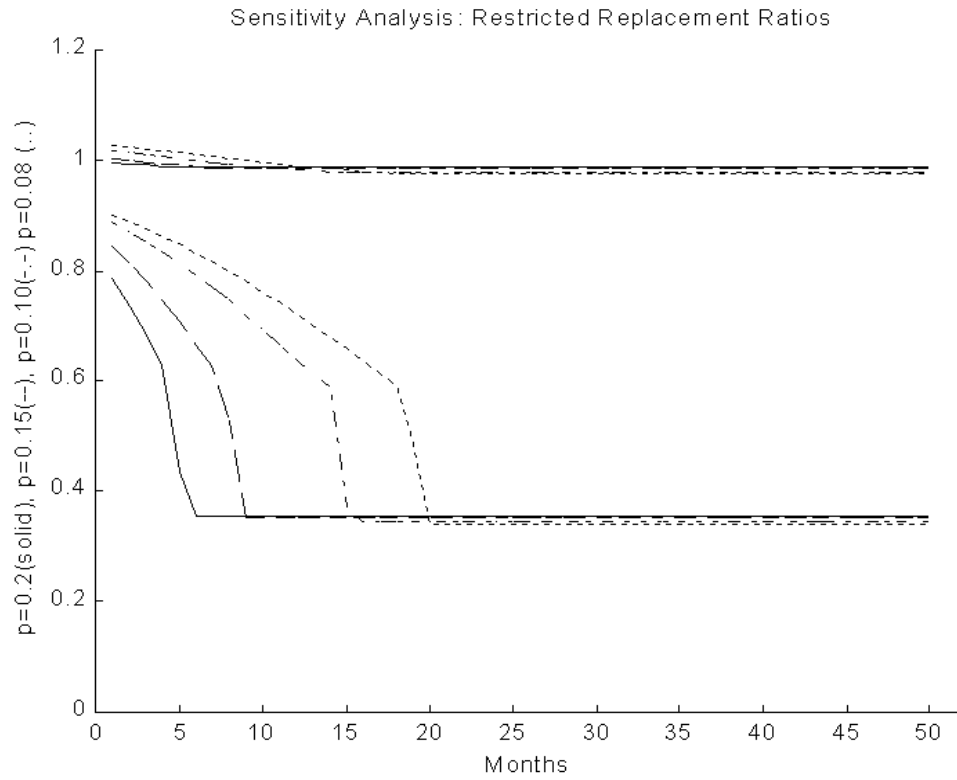


Figure 7: **Sensitivity analysis: different hazard rates.** In this figure we present the simulation results of the optimal unemployment insurance scheme for different hazard rate levels $p = 0.08, 0.1, 0.15, 0.2$. The horizontal axis represents unemployment duration (in months). The vertical axis represents replacement ratios.

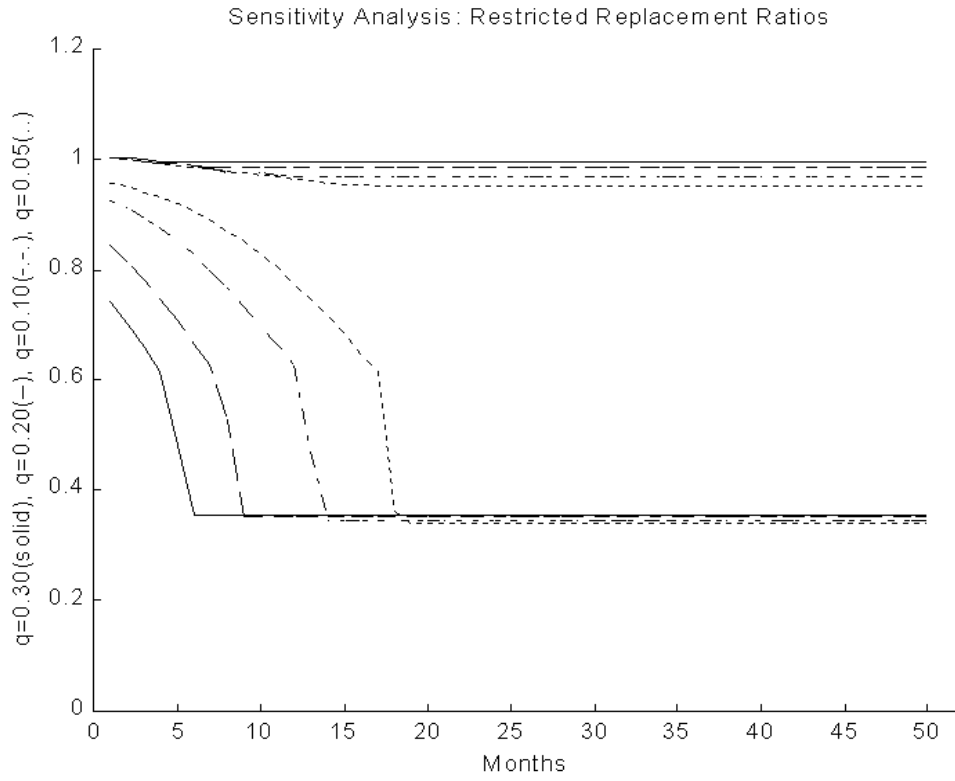


Figure 8: **Sensitivity analysis: different minimum utility bounds $U_{\min}$.** We vary the minimum bound by changing the value of the parameter q in equation (23). The horizontal axis represents unemployment duration (in months). The vertical axis represents replacement ratios.

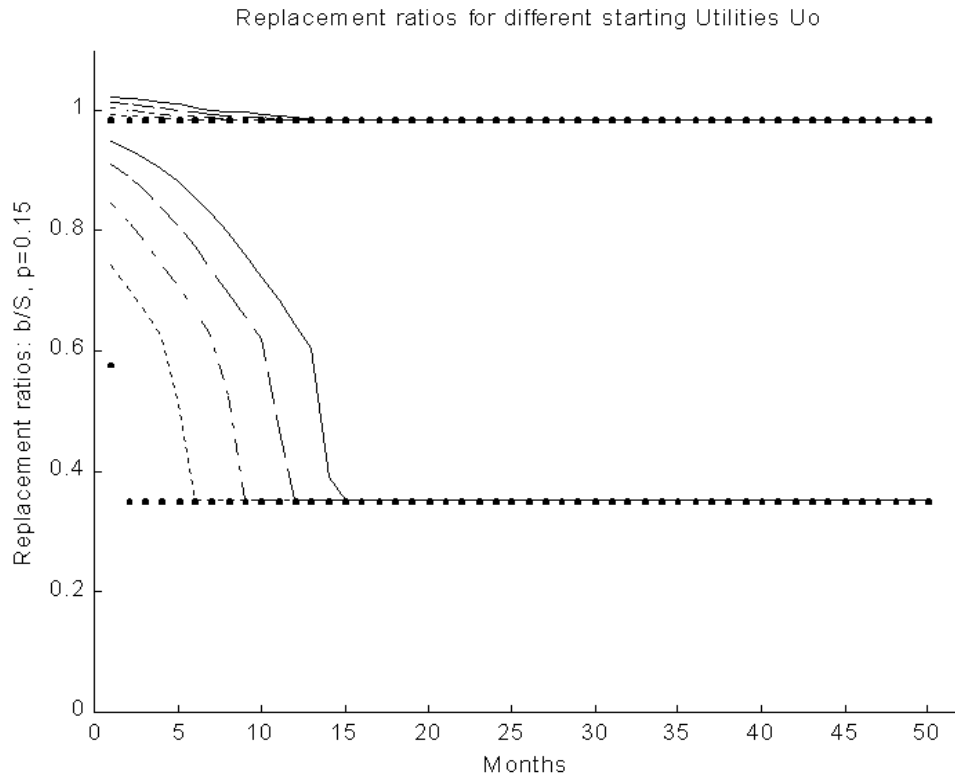


Figure 9: **Sensitivity analysis: different starting utilities U_0 .** The horizontal axis represents unemployment duration (in months). The vertical axis represents replacement ratios.

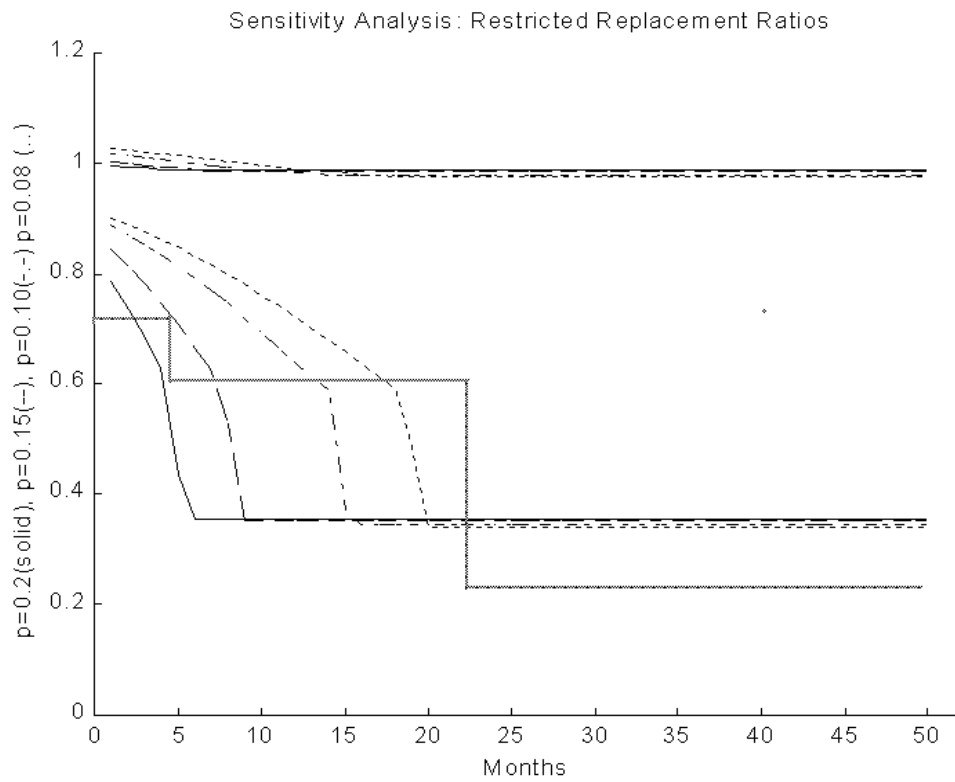


Figure 10: **Different p and comparison with the Spanish compensation scheme.** The thick solid step-shaped line reproduces the actual unemployment compensation scheme in Spain. The horizontal axis represents unemployment duration (in months). The vertical axis represents replacement ratios.

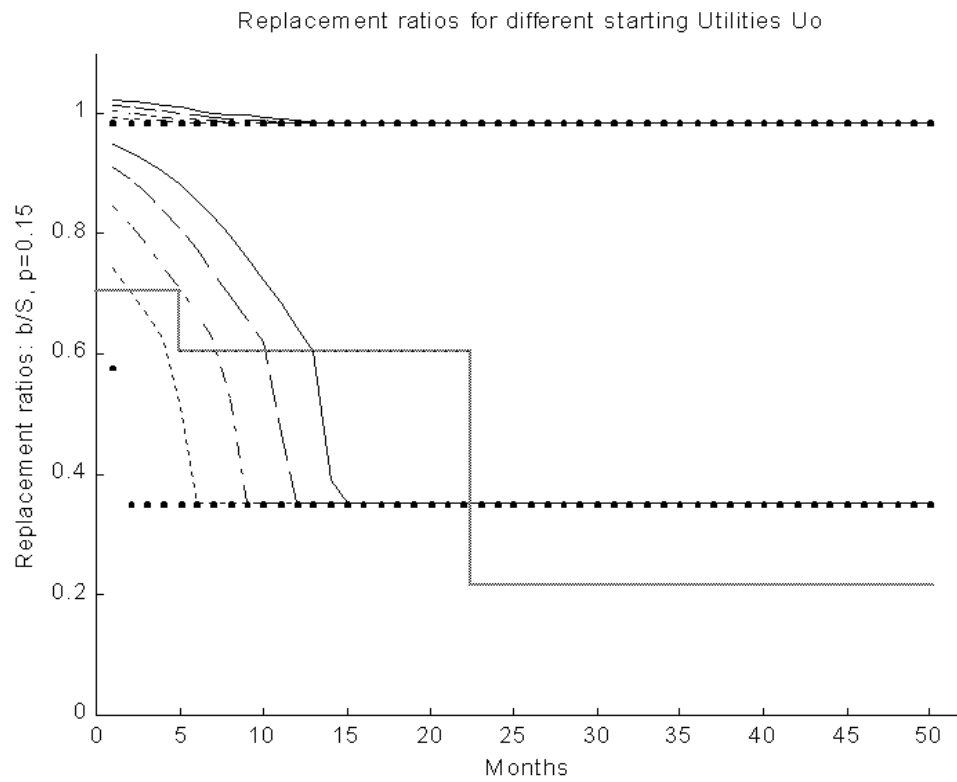


Figure 11: **Different U_0 and comparison with the Spanish compensation scheme.** The thick solid step-shaped line reproduces the actual unemployment compensation scheme in Spain. The horizontal axis represents unemployment duration (in months). The vertical axis represents replacement ratios.

Table 1 Selected features of unemployment benefit programmes

Country	Benefit type	Prior employment qualification	Fixed or earning related payment	Payment taxable ¹	Maximum duration of payment
Australia	UA	Nil	Fixed	Yes	Indef
Austria	UI	26 wks/12 mths; 52 wks/2 yrs if first claim, max benefit duration if 156 wks/3 yrs	ER	No, UI and UA set as % of after-tax income	30wk ²
	UA	Nil	ER		Indef
Belgium	UI	312 days/18 mths ³	ER ¹	Yes	Indef
Canada	UI	420-700 hrs/yr	ER	Yes	45 wks
	SW	Nil	Fixed		Indef
Czech Republic	UI	12 mths/3 yrs	ER	No	26 wks
Denmark	UI	26 wks/3 yrs	ER	Yes	5 yrs
Finland	UI	26 wks/2 yrs	ER	Yes	500 dys in 4 yrs
	UA	Nil	Fixed	Yes	Indef
France	UI	91 days/12 mths	ER ²	Yes	60 mths
	UA	5 yrs/10 yrs	Fixed		Indef
Germany	UI	360 days/3 yrs	ER	No	78-832 weekdays ²
	UA	150 days/yr or exhausted UI benefits	Fixed	No ⁴	Indef
Greece	UI	125 days/14 mths	ER ²	Yes	12 mths
	UA	60/days/2yrs	ER	Yes	3 mths
Hungary	UI	48 mths emp	ER ²	Yes	2 yrs
Iceland	UI	400hrs/12 mths	Fixed ¹¹	Yes	5 yrs
Ireland	UI	39 wks/12 mths	Fixed	Yes	15 mths
	UA	Nil	Fixed	No	Indef
Italy	UI ¹¹	1 yr/2 yr	ER	Yes	180 days
Japan	UI	6 mths/ 12 mths	ER ¹²	No	90-300 days ¹³
Korea	UI	6 mths	ER	No	30-210 days
Luxembourg	UI	6 mths/12 mths	ER ¹⁴	Yes	1 yr/2yrs ¹⁵
Netherlands	UI	26 wks/39 wks for basic benefit and 4 yrs/5 yrs for extended	ER	Yes	6 mths-4.5 yrs ¹⁶
	UA	3 yrs/5 yrs	Fixed	Yes	12 mths
New Zealand	UA	Nil	Fixed	Yes	Indef
Norway	UI	Prior earning requirement	ER	Yes	3 yrs
Poland	UI	180 days/yr ¹⁷	ER	Yes	9-24 mths ¹⁸
Portugal	UI	540 days/2 yrs	ER	No	10-30 mths ¹⁹
	UA	180 days/1 yr	Fixed	No	10-30mths ²⁰
Slovak Republic	UI	1 yr/3 yrs	ER		6 mths

Figure 12: **Table 1.** *Source* Kalisch, D. W., T. Aman and L. A. Buchele (1998), "Social and Health Policies in OECD Countries: A Survey of Current Programmes and Recent Developments," *Labour Market and Social Policy - Occasional Papers no. 33*, 67299 and 67364. Paris: OECD.

Table 1 Selected features of unemployment benefit programmes (continued)

Country	Benefit type	Prior employment qualification	Fixed or earning related payment	Payment taxable	Maximum duration of payment
Spain	UI	12 mths/6 yrs	ER	Yes	up to 24 mths ²
	UA	Exhausted UI or worked 6 mths	Fixed	Yes	6 - 18 mths
Sweden	UI	5 mths/12 mths	ER	Yes	300 days ³
	SW	Nil	Fixed	Yes	150-450 days ⁴
Switzerland	UI	6 mths during a base period of 2yrs	ER	Yes	150-400 days ⁵
United Kingdom	UI	1 yr/2 yrs ⁶	Fixed	Yes	26 wks
	GI	Nil	Fixed	Yes	Indef
United States ^a	UI	Yes ^b	ER	Yes	26 wks

Source: US Social Security Administration (1997), *Social Security Programs Throughout the World*, OECD(1998), *Benefits and Incentives in OECD countries 1995*, OECD *Caring World Synthesis*, Questionnaire responses, OECD *Economic Surveys* (various issues).

1. While payment may be treated as taxable income, tax scales may mean no tax is payable on this income.
2. In special cases, up to 52 weeks. If minimum employment contribution duration only 20 weeks.
3. Qualifying conditions rise with age of claimant up to 600 days in last 36 months.
4. Earnings related ratio declines after initial 12 months, with supplement for those with dependants, term fixed rate thereafter 30% replaceable when annual income exceeds 1.5 times maximum insurable earnings. Remainder taxable.
5. Earnings related amount declines incrementally as unemployment duration increases.
6. UI and UA set as proportion of after-tax earnings.
7. Varies according to insured employment period and age of recipient.
8. Supplement for dependants, and total amount subject to a 66.6% wage unskilled worker.
9. Earnings related declines after first yr., subject to national minimum/maximum amounts.
10. Amount increases according to period of prior employment, supplement for dependent children.
11. Other, more generous payments are also available for some without work, such as wage supplement/allowance payments, CIGG and CIGS, but they are not included here within the scope of unemployment benefits as the employment contract is generally not severed. Those displaced as a result of industry restructuring may also be eligible for a more generous mobility allowance.
12. Higher earnings replacement for prior lower income earners, subject to minimal/maximum limits.
13. Time increases with age, length of insurance, poor employment prospects.
14. Amount reduced if living with person whose wage exceeds 2 times social minimum wage.
15. Extension possible for further 6-12 months for hard-to-place and/or older unemployed.
16. Payment for up to 4 yrs requires contributions for 3 yrs in last 5 yrs.
17. No employment history requirement for those having completed studies, relieved from military service, completed maternity leave or released from prison.
18. Shorter period for young persons without employment history, period increases as prior employment history increases and/or firm bankrupt. 12 months is the norm for most workers (except young, women with 25 years prior employment and men with 30 years employment).
19. Maximum duration of payment depends on age of recipient.
20. Payment duration increases with prior contribution history.
21. Payable up to 450 days if age 55-64 for UI; for SW benefit payable up to 150 days if below age 55, 300 days if age 55-59 and 450 days if age 60-64.
22. Maximum payment period for passive benefits of 150 days if under age 50, 250 days if age 50-60 and 400 days if over age 60. Can be extended to 520 days in total for all through participation in active labour market programmes.
23. Special earnings related contribution history required, rather than just employment qualification.
24. States have own laws, some differences between schemes.
25. About 3/4 of States have minimum earnings requirement over last year, remainder require employment of approx. 15-20 wks in last yr.

Figure 13: **Table 1.** Source Kalisch, D. W., T. Aman and L. A. Buchele (1998), "Social and Health Policies in OECD Countries: A Survey of Current Programmes and Recent Developments," *Labour Market and Social Policy - Occasional Papers no. 33*, 67299 and 67364. Paris: OECD.